

Replay-Aware TSP Benchmark Report

Overview

- Condition count: 9.
- Best final transfer gap: phase6_failure_replay.
- Best TSPLIB holdout gap: phase6_failure_replay.
- Best synthetic holdout gap: phase6_random_replay.

Run Metadata

- run_name: run_20260518_223014_a
- started_at_local: 2026-05-18 22:30:14
- finished_at_local: 2026-05-18 22:51:56
- duration_hhmm: 00:22
- duration_seconds: 1302.613
- seed_offset: 0
- replicate_label: a
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Archive Diversity	Archive Hardness	Failure Concentration	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
phase6_no_replay	none	score_only	False	0.213387	0.064916	0.159398	0.236772	0.214847	0.0	0.869493	0.56	-0.028643
phase6_random_replay	random	score_only	False	0.124445	0.0	0.079192	0.236772	0.143152	0.19457	0.692413	0.76	0.052041
phase6_failure_replay	failure	score_only	False	0.082632	0.0	0.052584	0.171206	0.258808	0.191132	0.794418	0.76	0.058353

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Archive Diversity	Archive Hardness	Failure Concentration	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
phase6_random_replay_compression	random	novelty_gate	True	0.170887	0.010343	0.112507	0.236772	0.1693	0.19457	0.0	0.76	0.0
phase6_stratified_random_replay	stratified_random	score_only	False	0.132743	0.010343	0.088234	0.236772	0.164957	0.221162	0.815976	0.76	0.014133
phase6_diversity_weighted_replay	diversity_weighted	score_only	False	0.213387	0.064916	0.159398	0.236772	0.214847	0.130039	0.901121	0.56	-0.027638
phase6_residual_failure_replay	residual_failure	score_only	False	0.213387	0.064916	0.159398	0.149274	0.338865	0.291666	0.812712	0.56	-0.030644
phase6_diversity_failure_replay	diversity_failure	score_only	False	0.124231	0.010343	0.082817	0.171206	0.272545	0.173881	0.0	0.76	0.0
phase6_diversity_failure_replay_compression	diversity_failure	novelty_gate	True	0.09781	0.0	0.062243	0.171206	0.261764	0.17024	0.78376	0.76	0.039306

Condition Notes

phase6_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.869493, and final complexity 0.56.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.214847, size bias 8.3e-05, and failure concentration 0.0.
- Replay selection diversity 0.0, mean selected expected gap 0.0, mean selected residual gap 0.0.
- Adaptation efficiency -0.028643 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.399069, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": 0.002249}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.257552, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.133545}],

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{"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.321274, "family": "lin", "name": "lin105",  
"optimality_gap": 0.328813, "residual_gap": 0.007539}]
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phase6_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.124445, synthetic 0.0, combined 0.079192.
- Accepted-epoch count 2, mean accepted code novelty 0.692413, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.143152, size bias 8.3e-05, and failure concentration 0.19457.
- Replay selection diversity 0.262174, mean selected expected gap 0.133447, mean selected residual gap -0.022931.
- Adaptation efficiency 0.052041 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 2, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.124445 across 7 instances; family means: ch=0.148214, kroD=0.176341, pcb=0.184785, pr=0.057702, rd=0.102528, st=0.053333.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3459, "expected_gap": 0.402714, "family": "a", "name": "a280", "optimality_gap": 0.341218, "residual_gap": -0.061496}, {"best_known_cost": 14379, "cost": 18571, "expected_gap": 0.323194, "family": "lin", "name": "lin105", "optimality_gap": 0.291536, "residual_gap": -0.031658}, {"best_known_cost": 629, "cost": 799, "expected_gap": 0.25628, "family": "eil", "name": "eil101", "optimality_gap": 0.27027, "residual_gap": 0.01399}]

phase6_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.082632, synthetic 0.0, combined 0.052584.
- Accepted-epoch count 2, mean accepted code novelty 0.794418, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.258808, size bias 0.120322, and failure concentration 0.191132.
- Replay selection diversity 0.212069, mean selected expected gap 0.308948, mean selected residual gap -0.006576.
- Adaptation efficiency 0.058353 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.082632 across 7 instances; family means: ch=0.069889, kroD=0.033765, pcb=0.205522, pr=0.044546, rd=0.1, st=0.054815.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3658, "expected_gap": 0.404808, "family": "a", "name": "a280", "optimality_gap": 0.418379, "residual_gap": 0.013571}, {"best_known_cost": 629, "cost": 870, "expected_gap": 0.25469, "family": "eil", "name": "eil101", "optimality_gap": 0.383148, "residual_gap": 0.128458}, {"best_known_cost": 14379, "cost": 18416, "expected_gap": 0.322373, "family": "lin", "name": "lin105", "optimality_gap": 0.280757, "residual_gap": -0.041616}]

phase6_random_replay_compression

- Replay mode: random with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.170887, synthetic 0.010343, combined 0.112507.

- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.1693, size bias 8.3e-05, and failure concentration 0.19457.
- Replay selection diversity 0.262174, mean selected expected gap 0.13189, mean selected residual gap 0.002319.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.170887 across 7 instances; family means: ch=0.167174, kroD=0.250305, pcb=0.20434, pr=0.21768, rd=0.164349, st=0.025185.
- Panel synthetic_holdout mean gap 0.010343 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.041372, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 14379, "cost": 19824, "expected_gap": 0.321803, "family": "lin", "name": "lin105", "optimality_gap": 0.378677, "residual_gap": 0.056874}, {"best_known_cost": 2579, "cost": 3361, "expected_gap": 0.400776, "family": "a", "name": "a280", "optimality_gap": 0.303218, "residual_gap": -0.097558}, {"best_known_cost": 629, "cost": 804, "expected_gap": 0.253418, "family": "eil", "name": "eil101", "optimality_gap": 0.278219, "residual_gap": 0.024801}]

phase6_stratified_random_replay

- Replay mode: stratified_random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.132743, synthetic 0.010343, combined 0.088234.
- Accepted-epoch count 3, mean accepted code novelty 0.815976, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.164957, size bias 8.3e-05, and failure concentration 0.221162.
- Replay selection diversity 0.187367, mean selected expected gap 0.236142, mean selected residual gap -0.039676.
- Adaptation efficiency 0.014133 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.132743 across 7 instances; family means: ch=0.086816, kroD=0.08021, pcb=0.244003, pr=0.272303, rd=0.070164, st=0.088889.
- Panel synthetic_holdout mean gap 0.010343 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.041372, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3535, "expected_gap": 0.40031, "family": "a", "name": "a280", "optimality_gap": 0.370686, "residual_gap": -0.029624}, {"best_known_cost": 7542, "cost": 9490, "expected_gap": 0.235826, "family": "berlin", "name": "berlin52", "optimality_gap": 0.258287, "residual_gap": 0.022461}, {"best_known_cost": 14379, "cost": 17716, "expected_gap": 0.321511, "family": "lin", "name": "lin105", "optimality_gap": 0.232075, "residual_gap": -0.089436}]

phase6_diversity_weighted_replay

- Replay mode: diversity_weighted with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.901121, and final complexity 0.56.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.214847, size bias 8.3e-05, and failure concentration 0.130039.
- Replay selection diversity 0.266358, mean selected expected gap 0.167625, mean selected residual gap 0.026065.

- Adaptation efficiency -0.027638 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.400931, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": 0.000387}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.254054, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.137043}, {"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.321511, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": 0.007302}]

phase6_residual_failure_replay

- Replay mode: residual_failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.812712, and final complexity 0.56.
- Active replay archive residual_archive with mean diversity 0.149274, mean hardness 0.338865, size bias 0.093558, and failure concentration 0.291666.
- Replay selection diversity 0.177489, mean selected expected gap 0.265344, mean selected residual gap 0.087049.
- Adaptation efficiency -0.030644 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.402869, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": -0.001551}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.262003, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.129094}, {"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.321274, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": 0.007539}]

phase6_diversity_failure_replay

- Replay mode: diversity_failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.124231, synthetic 0.010343, combined 0.082817.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.272545, size bias 0.120322, and failure concentration 0.173881.
- Replay selection diversity 0.193082, mean selected expected gap 0.310392, mean selected residual gap -0.015991.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.124231 across 7 instances; family means: ch=0.044076, kroD=0.256035, pcb=0.189866, pr=0.208101, rd=0.102276, st=0.025185.
- Panel synthetic_holdout mean gap 0.010343 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.041372, ring_bridge=0.0, two_corridors=0.0.

- Worst recent training cases: [{"best_known_cost": 14379, "cost": 19562, "expected_gap": 0.317296, "family": "lin", "name": "lin105", "optimality_gap": 0.360456, "residual_gap": 0.04316}, {"best_known_cost": 2579, "cost": 3391, "expected_gap": 0.403257, "family": "a", "name": "a280", "optimality_gap": 0.314851, "residual_gap": -0.088406}, {"best_known_cost": 629, "cost": 760, "expected_gap": 0.267091, "family": "eil", "name": "eil101", "optimality_gap": 0.208267, "residual_gap": -0.058824}]

phase6_diversity_failure_replay_compression

- Replay mode: diversity_failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.09781, synthetic 0.0, combined 0.062243.
- Accepted-epoch count 3, mean accepted code novelty 0.78376, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.261764, size bias 0.120322, and failure concentration 0.17024.
- Replay selection diversity 0.198286, mean selected expected gap 0.301127, mean selected residual gap -0.042301.
- Adaptation efficiency 0.039306 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 2, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.09781 across 7 instances; family means: ch=0.074095, kroD=0.107636, pcb=0.183465, pr=0.088028, rd=0.081795, st=0.075556.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3259, "expected_gap": 0.40349, "family": "a", "name": "a280", "optimality_gap": 0.263668, "residual_gap": -0.139822}, {"best_known_cost": 629, "cost": 686, "expected_gap": 0.263911, "family": "eil", "name": "eil101", "optimality_gap": 0.09062, "residual_gap": -0.173291}, {"best_known_cost": 14379, "cost": 15443, "expected_gap": 0.328729, "family": "lin", "name": "lin105", "optimality_gap": 0.073997, "residual_gap": -0.254732}]

Judge Appendix

Summary and Interpretation of Replay-Aware TSP Benchmark Results

General Notes

- Primary transfer evidence is taken from **held-out TSPLIB gap** and **synthetic holdout gap** metrics.
- Replay mechanism studies focus on archive diversity, hardness, size bias, failure concentration, and failure spread.
- Code novelty and transfer are explicitly contrasted.
- Replay policies distinguish between broad-coverage (random, stratified random), raw-failure, residual-failure, diversity-weighted, and compression-aware replay.

Top Performing Conditions

Condition	Ts pli b Ga p	Synt heti c Gap	Tra nsf er Gap	Replay Mode	Code Novelty (mean)	Archiv e Diver sity	Archi ve Har dness	Failure C oncentra tion	Notes
phase6_failure_replay	**0.0826**	0.0	**0.0526**	failure	0.7944	0.1712	0.2588	0.2245	Best transfer, lowest gaps
phase6_diversity_failure_replay_compression	0.0978	0.0	0.0622	diversity_failure with compression	0.7838	0.1712	0.2618	0.1702	Good transfer, compression applied
phase6_random_replay	0.1244	0.0	0.0792	random	0.6924	0.2368	0.1432	0.1946	Moderate transfer, higher diversity
phase6_stratified_random_replay	0.1327	0.0103	0.0882	stratified_random	0.8160	0.2368	0.1650	0.2212	Slightly worse transfer
phase6_diversity_failure_replay	0.1242	0.0103	0.0828	diversity_failure	0.0000	0.1712	0.2725	0.1739	Code novelty dropped to zero
phase6_no_replay	0.2134	0.0649	0.1594	none	0.8695	0.2368	0.2148	0.0	Worst transfer, high code novelty

****Notes:****

- ****Best transfer**** performance is from ****phase6_failure_replay**** with lowest heldout TSPLIB gap (0.0826) and transfer gap (0.0526).
- ****Random replay variants**** (including compression-aware) yield intermediate transfer improvements, with varying code novelty.
- ****No replay**** baseline shows largest transfer gaps despite highest code novelty and balanced behavior.
- Replay modes with ****diversity or failure weighting plus compression**** improve transfer gaps relative to random replay without compression.
- ****Code novelty is not strongly correlated with better transfer:**** the ****diversity_failure_replay**** condition has zero code novelty yet shows competitive transfer gaps.
- Failure replay modes tend to have ****higher archive hardness and failure concentration**** than random modes.

Replay Archive Characteristics by Condition

Condition	Ar ch. Siz e	Arch. Diver sity	Arch. Hard ness	Arch. Size Bias	Failure Conc. Final	Failure Conc. Mean	Replay Selection Diversity	Comments
phase6_failure_replay	31	0.1712	0.2588	0.1203	0.2245	0.1911	0.212	Concentrated replay on failures, moderate diversity

Condition	Arch. Size	Arch. Diversity	Arch. Hardness	Arch. Size Bias	Failure Conc. Final	Failure Conc. Mean	Replay Selection Diversity	Comments
phase6_diversity_failure_replay_compression	28	0.1747	0.2878	0.1250	0.1702	0.1702	0.1983	Compression modestly reduces failure concentration
phase6_random_replay	28	0.2368	0.1432	0.0000	0.1946	0.1946	0.262	Highest diversity, lowest hardness
phase6_stratified_random_replay	30	0.2368	0.1650	0.0000	0.2212	0.2212	0.187	Balanced characteristics
phase6_diversity_failure_replay	30	0.1747	0.3198	0.1250	0.1739	0.1739	0.193	Lower diversity, higher hardness
phase6_no_replay	32	0.2368	0.2148	0.0	0.0	0.0	0.0	No replay, no failure concentration
phase6_residual_failure_replay	31	0.1497	0.3386	0.0944	0.3333	0.2917	0.177	High failure concentration and hardness (residual selected)
phase6_diversity_weighted_replay	32	0.2402	0.2124	0.0	0.1300	0.1300	0.266	Largest diversity, moderate failure concentration

****Insights:****

- Failure-based replay (failure, residual_failure, diversity_failure) tends to yield ****higher failure concentrations**** and ****hardness**** in archives than random or no replay.
- Diversity weighting supports ****higher descriptor diversity**** but may come at cost of failure concentration dilution.
- Compression pressure cases (random_replay_compression, diversity_failure_replay_compression) have ****some reduction**** in failure concentration but maintain decent transfer gap.
- No replay generates ****no failure concentration**** or replay diversity (as expected).

Transfer and Optimality Gap Trends

- ****Transfer gaps on held-out TSPLIB sets:**** lowest for failure replay (0.0826), followed by diversity failure with compression (0.0978), then random replay (0.1244), and worst for no replay (0.2134).
- ****Synthetic holdout gaps:**** minimal (zero or near zero) for replay-enabled conditions; higher only for no replay (0.0649).
- ****Across all best-performing replay conditions, the transfer gap is substantially better than no replay****, indicating benefits from replay interventions.
- ****Failure replay usage strongly correlates with best transfer gaps and lowest optimality gaps on held-out TSPLIB instances.****

Code Novelty vs Transfer

- Conditions with ****high code novelty (~0.8-0.9)**** include no replay and diversity weighted replay, but they exhibit ****worse or equal transfer gaps**** compared to failure replay modes which have ****moderate to low novelty**

(~0.69-0.79)**.

- Some diversity or highest novelty replay conditions (e.g., "phase6_diversity_failure_replay") show **zero mean code novelty**, yet maintain decent transfer gaps.
- This suggests **code novelty metrics do not directly imply algorithmic improvement or transfer efficacy** in this setting.

Summary Conclusions

- **Failure-replay mechanisms (phase6_failure_replay) achieve the best transfer to held-out TSPLIB and synthetic benchmarks, with lowest gaps and reasonable archive diversity and hardness.**
- **Random replay with no selection or simple random replay yields moderate transfer but somewhat higher gaps and lower hardness.**
- **Increased archive hardness and replay failure concentration correlate with improved transfer performance.**
- **Compression-aware replay situates between random and failure replay in terms of transfer, maintaining archive structure with moderate failure concentration.**
- **Code novelty does not track transfer improvement, sometimes falling when transfer improves (diversity_failure_replay). Thus, code novelty is not a proxy for meaningful algorithmic invention here.**
- **Replay strategy impacts archive characteristics substantially: failure replay concentrates on hard failures but may reduce diversity; diversity-weighted replay maximizes diversity but with a possible cost in failure concentration.**
- Future investigations should analyze method adaptation efficiency in conjunction with replay modes and failure concentration to optimize transfer.

Recommendations

- Prioritize **failure replay modes** for best transfer and generalization to TSPLIB and synthetic test sets.
- Consider combining diversity weighting with failure replay to balance diversity and hardness without sacrificing transfer.
- Monitor code novelty separately and avoid equating it with algorithmic invention or transfer gains.
- Archive size bias and complexity seem stable and less informative for transfer predictions; focus on failure concentration and hardness instead.
- Evaluate compression effects as a control to maintain archive manageability without harming transfer.

End of analysis.